PHASE SHIFT

Factoring B out of the argument

The shift of $y = \sin(2x - \pi)$ is $\frac{\pi}{2}$, not π. The factored form makes that visible.

BEFORE

General form

$$y = \sin(2x - \pi)$$

Reading $C = \pi$ as the shift is the common error

FACTOR OUT B

take $B = 2$ out of the argument

AFTER

Factored form

$$y = \sin\left(2\left(x - \frac{\pi}{2}\right)\right)$$

Shift $\frac{\pi}{2}$ to the right, read straight off

WHAT CHANGED

The argument becomes B times a bracket, and the shift $\frac{C}{B}$ sits inside that bracket.

WHAT STAYED

The function and its graph: the standard cycle still starts where the argument is zero, at $x = \frac{\pi}{2}$.